PS2 - Motion Planning

IVDC Autonomy 2026 | Research and learning answers

Section 1 - Path-planning basics

1. What is the difference between a path and a trajectory?

A path is a list of places to visit. A trajectory also says when to reach them and may include speed. This program finds a path on a grid.

2. What do waypoint, obstacle, free space, start state, and goal mean?

- A waypoint is a place along the route.
- An obstacle is an area to avoid.
- Free space is an area the vehicle can use.
- The start state is where the trip begins.
- The goal is where the trip should end.

Section 2 - Map representation

3. What is a 2D grid map?

A grid map divides an area into rows and columns. Each cell in this program is free or blocked. The search can move one cell up, down, left, or right.

4. What is an occupancy grid?

An occupancy grid records whether places are free, blocked, or not yet known. A robot can update its map from sensor readings. This program uses a fixed map.

5. What are free, occupied, and unknown cells?

A free cell can be used. An occupied cell has an obstacle. An unknown cell has not been checked yet. The example map has free and occupied cells.

Section 3 - Search algorithms

6. How does Breadth-First Search (BFS) work?

BFS uses a first-in, first-out queue. It checks nearby cells first, then cells farther away. It marks a cell visited when it is added to the queue. If every move has the same cost, BFS finds a path with the fewest moves.

7. How does Dijkstra's Algorithm work, and how is it different from BFS?

Dijkstra checks the cell with the lowest known travel cost first. It can handle different non-negative move costs. BFS is simpler when every move costs the same.

8. How does A* use a heuristic?

A* gives each cell a score: f = g + h. g is the cost from the start. h estimates the cost to the goal. On a four-direction grid, Manhattan distance is a simple estimate. A* can find a shortest path when the estimate does not overstate the remaining cost.

9. How are BFS, Dijkstra, and A* different?

BFS is for equal-cost moves. Dijkstra handles different move costs. A* also uses an estimate to guide its search toward the goal. With equal-cost grid moves, BFS and A* can both find a shortest path.

Section 4 - Global and local planning

10. What is the difference between a global path and a local motion decision?

A global path gives a route across a map. A local decision chooses what to do nearby, based on the space around the vehicle. This program makes a basic global path.

11. Why might a vehicle need to re-plan?

A new obstacle may block the route. The vehicle can update its map and search for another path. It must still check that the new path is safe to follow.

References

- Amit Patel, Red Blob Games, Introduction to the A* Algorithm: https://www.redblobgames.com/pathfinding/a-star/introduction.html.
- Amit Patel, Red Blob Games, A* Implementation: https://www.redblobgames.com/pathfinding/a-star/implementation.html.